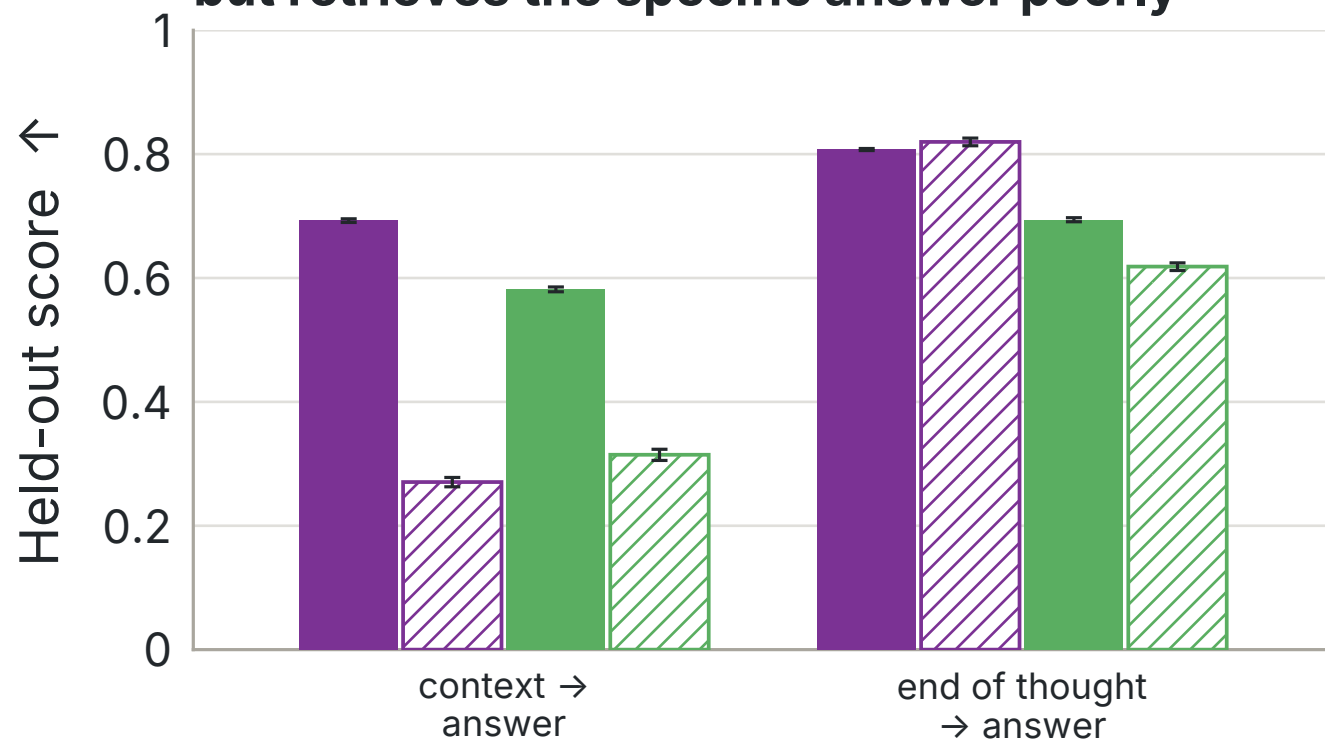


CORPORA

Needs-reasoning corpora No-reasoning corpora

A · OPENTHINKER3-7B, LAYER 19

**The context state predicts the answer state,
but retrieves the specific answer poorly**

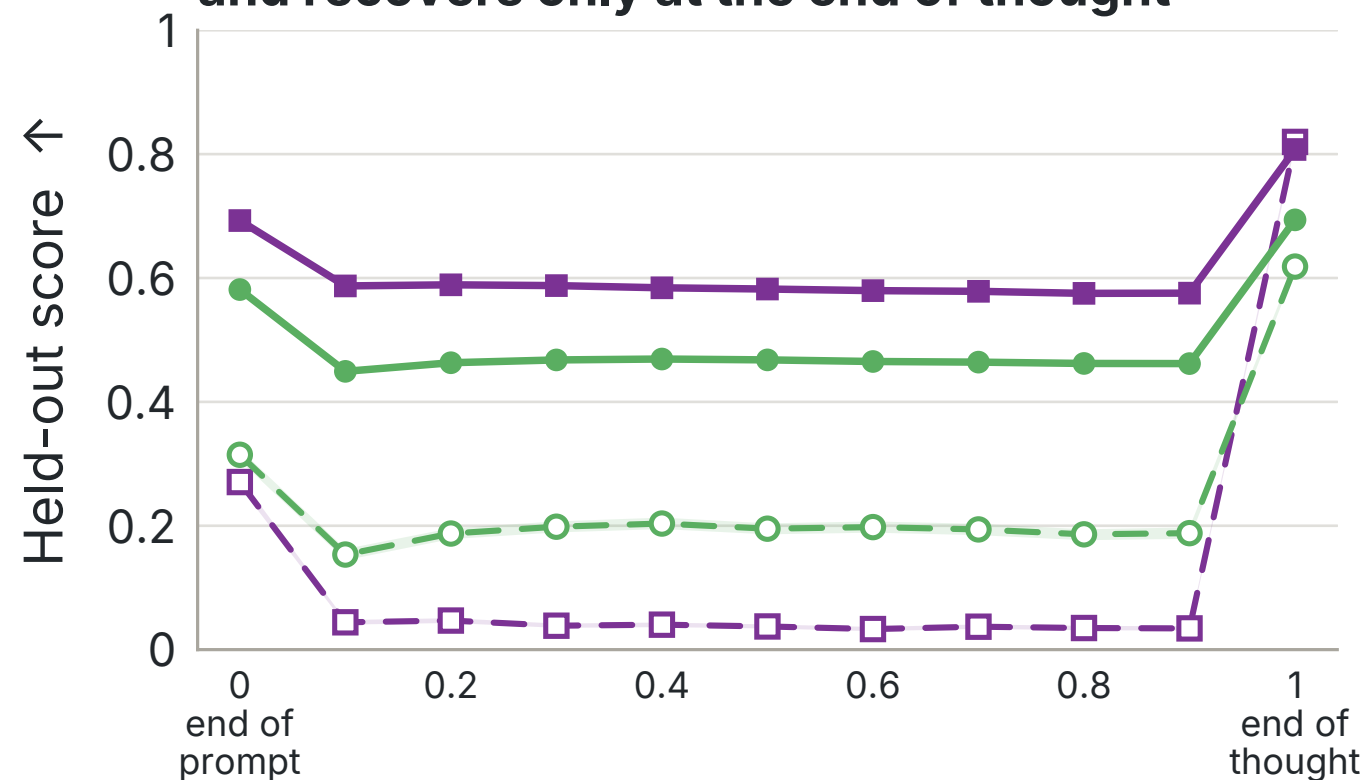


METRIC

Held-out R^2 Answer retrieval (lift over chance)

B · OPENTHINKER3-7B, LAYER 19

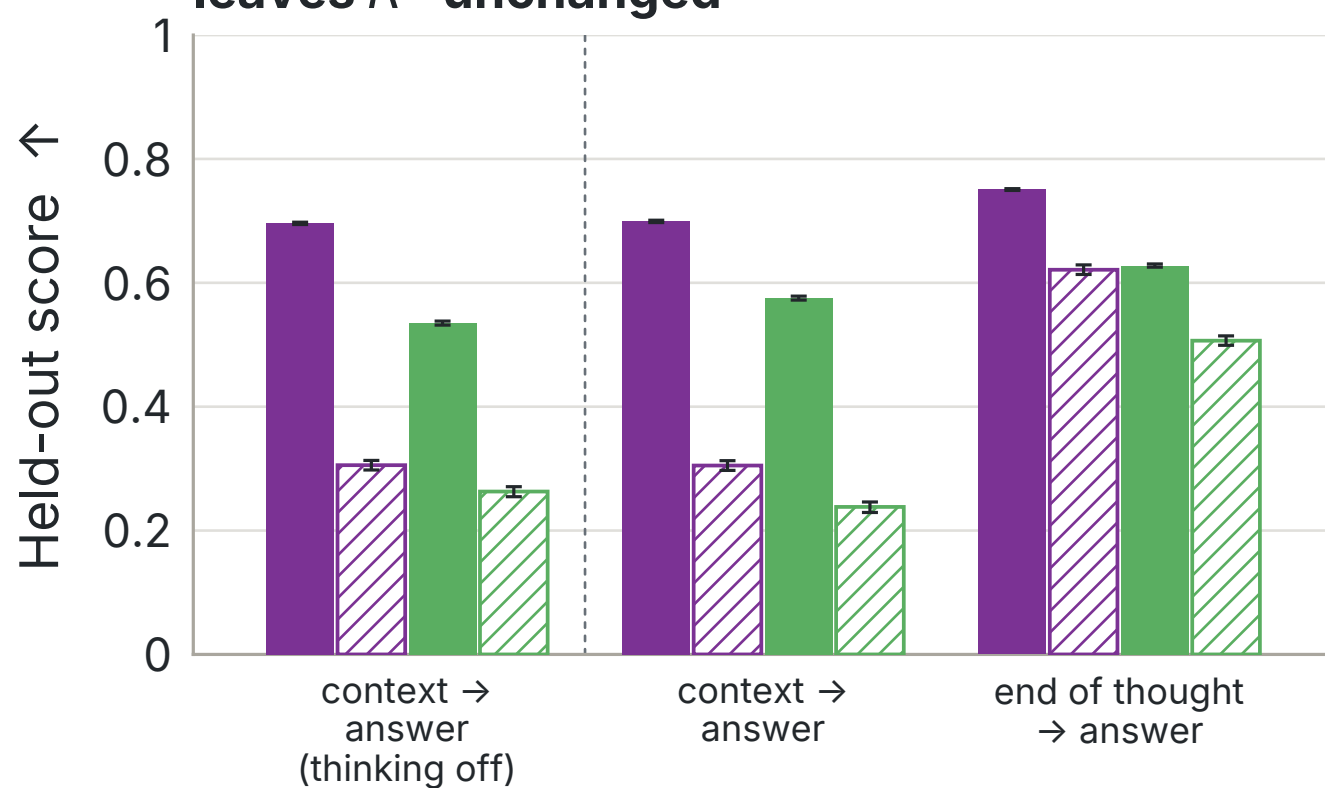
**Predictability dips inside the reasoning trace
and recovers only at the end of thought**



Position in the thinking span

C · QWEN3-8B, LAYER 24

**Disabling thinking on the same weights
leaves R^2 unchanged**



D · QWEN2.5-7B-INSTRUCT (BEFORE) → OPENTHINKER3-7B (AFTER), LAYER 19

**The context state before reasoning training
predicts the answer state after it**

